

Technical Glossary

A comprehensive glossary of technical terms, acronyms, and concepts used in the Archimedes Reference Architecture. Navigate by clicking on letters below or scroll to browse all terms.

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A B C D E H I J L M P R S T U
X

A

AMP (Analytics Management Platform)

A core system component that sends structured data messages to one or more DAS instances using HTTP REST protocols. AMP acts as the data source in the test environment, transmitting operational status, test results, and system events to configured DAS endpoints at regular intervals defined by heartbeat settings.

SYSTEM COMPONENTS

Adaptive Test

A testing methodology where test parameters, limits, or execution flow can be dynamically modified during runtime based on intermediate results or external commands. Adaptive DAS systems enable real-time control of test programs to optimize quality and efficiency.

TESTING

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B B

BIN (Binning)

Classification category for tested devices based on test results. Devices are sorted into bins representing pass/fail status and quality grades. Each bin typically has a numeric identifier and optional name describing the classification criteria.

TESTING

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C C

CZ (Characterization)

A specialized test mode focused on collecting detailed performance data across various operating conditions. CZ messages include CZ_SETUP, CZ_START, CZ_DATA_POINT, CZ_SUMMARY, and CZ_END to track characterization test execution and results.

TESTING

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DAS (Data Analytics Solution)

A system designed to receive, process, and manage structured data sent by test applications such as AMP. Each DAS instance exposes REST endpoints to accept HTTP messages, validate their content, and respond with subscription lists. DAS systems enable real-time data collection, analysis, and storage of test operations.

SYSTEM COMPONENTS

DAS Listener

A REST-based server component that implements the DAS architecture. It listens on HTTP endpoints, receives POST requests containing JSON payloads, validates message structure and sequence, and returns appropriate HTTP status codes and response payloads.

SYSTEM COMPONENTS

DAS_Conf.xml

Configuration file that defines all DAS instances used by AMP. Specifies endpoint URLs, message queue limits (MaxMsg), heartbeat intervals, and enable/disable status for each DAS connection. This file controls how AMP distributes messages across multiple DAS services.

CONFIGURATION

DTR (Datalog Test Record)

Detailed test result data containing individual parametric measurements, test numbers, limits, and pass/fail status. DTR messages provide comprehensive test execution information for analysis and traceability.

MESSAGES

E E

Endpoint

A specific URL path on a REST server that accepts HTTP requests. DAS endpoints follow patterns like /TESTCELL/{hostname}/{messagename} or include additional routing parameters for LOT_ID and SUBLOT_ID to organize message flow.

ARCHITECTURE

H H

Heartbeat

The interval (in seconds) at which AMP automatically sends periodic messages to all enabled DAS instances. Heartbeat ensures continuous communication, synchronization, and early detection of connection issues. Typical values range from 20 to 60 seconds.

CONFIGURATION

HTTP (Hypertext Transfer Protocol)

The foundational protocol for data communication on the web. In DAS architecture, HTTP is used to transport JSON messages between AMP and DAS using methods like POST (send data) and GET (retrieve data).

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I I

IGXL (IG-XL Test Platform)

Teradyne's test platform for semiconductor testing. IGXL_DATA messages contain platform-specific operational data and test results formatted according to IG-XL specifications.

PLATFORMS

INITIALIZATION

The mandatory first message that AMP must send when establishing communication with a DAS. Contains system information, supported message types (TCS_MESSAGE_LIST), vendor details, and configuration parameters. If any other message is sent first, DAS must reject it with HTTP 500.

MESSAGES

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J J

JSON (JavaScript Object Notation)

A lightweight, human-readable data interchange format consisting of key-value pairs. Used as the standard message format for all AMP-DAS communication. JSON supports strings, numbers, booleans, arrays, and nested objects.

DATA FORMATS

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L L

LOT_ID

Unique identifier for a batch of devices being tested together. Used in message routing and data organization to track test results at the lot level. LOT_START and LOT_END messages mark lot boundaries.

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M M

MESSAGE_LIST

A JSON array returned by DAS in response to valid messages, containing the list of message types the DAS subscribes to. This tells AMP which message types this particular DAS wants to receive, enabling selective message distribution.

MESSAGES

MST (Mixed Signal Test)

Test platform and data format for mixed-signal semiconductor testing. MST_DATA messages contain test results and measurements specific to analog and mixed-signal device testing.

PLATFORMS

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P P

POST

HTTP method used to send data to a server. All AMP-to-DAS communication uses POST requests with JSON payloads in the request body. The server processes the data and returns a response with status code and optional payload.

PROTOCOLS

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R R

REST (Representational State Transfer)

An architectural style for distributed systems that uses standard HTTP methods to communicate. REST APIs expose endpoints (URLs) that clients can call to exchange data, typically using JSON format. DAS architecture is built on REST principles.

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S S

STATUS

A regular message type sent by AMP to report system operational state. Contains current user, software versions, tester status, and operational mode. Sent at intervals defined by the heartbeat configuration.

MESSAGES

SUBLOT_ID

Identifier for a subset of devices within a LOT. Enables finer-grained organization of test data. SUBLOT_START and SUBLOT_END messages mark subplot boundaries within a lot execution.

TESTING

SHUTDOWN

Message sent by AMP to notify DAS that communication is ending gracefully. Allows DAS to perform cleanup operations, finalize logs, and close resources properly before disconnection.

MESSAGES

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TCS (Test Cell Services)

Framework and protocol specifications for test cell communication. TCS defines message formats, communication patterns, and behavioral requirements. TCS_MESSAGE_LIST, TCS_VENDOR, and TCS_VERSION are key fields in INITIALIZATION messages.

PROTOCOLS

TEST_CELL

A logical grouping representing a test station or test system instance. Used as a routing parameter in DAS endpoint URLs to organize messages by physical or logical test location.

TESTING

TEST_DATA

Message containing parametric test results including measurements, pass/fail status, bin assignments, and test execution metadata. Core message type for transmitting actual test outcomes from AMP to DAS.

MESSAGES

TEST_START

Message indicating the beginning of test execution for a device or unit. Contains test program information, device identifiers, and test configuration parameters.

MESSAGES

TEST_END

Message marking completion of test execution for a device. Includes final bin assignment, pass/fail status, and summary information about the completed test.

MESSAGES

TIME_STAMP

Required field in all JSON messages containing an ISO 8601 formatted UTC timestamp. Ensures temporal ordering and traceability of all messages in the system. Format: "YYYY-MM-DDTHH:MM:SS.FFFFFFFZ"

DATA FORMATS

T_STATUS

Enhanced status message providing detailed operational state during testing. Has its own heartbeat interval configuration and includes extended information about active test execution.

MESSAGES

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U U

UltraEdge

Platform designed to host DAS applications and provide test cell services. UltraEdge systems run DAS listeners and adaptive test applications in the test environment.

PLATFORMS

URL (Uniform Resource Locator)

Address specifying the location of a resource on a network. DAS URLs identify specific endpoints where messages should be sent, including protocol (http/https), host, port, and path components.

PROTOCOLS

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X X

XML (Extensible Markup Language)

Structured markup language used for configuration files such as DAS_Conf.xml. Defines hierarchical data using tags and attributes. While DAS messages use JSON, configuration files use XML format.

DATA FORMATS

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Note: This glossary is maintained as part of the Teradyne Archimedes Reference Architecture documentation. For detailed information on any term, please refer to the specific documentation sections linked throughout this site.